Agent testing framework

Problem Statement

Task:

Build a **framework that can test any AI agent** (not just your own) using:

- Predefined test cases
- Automated evaluation
- Adversarial testing

Context

Most AI systems fail not because of poor models—but due to:

- Lack of testing
- No guardrail validation
- No evaluation pipeline

Your job is to fix that.

Requirements

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1. Agent-Agnostic Design

Your framework must:

- Accept any agent via a simple interface:

```
def run_agent(input: str) -> str:
  ...
```

This ensures:

- Reusability

- Standardization

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2. Test Case System

Must support:

- Structured test cases:

```
{
 "input": "User query",
 "expected_behavior": "What should happen",
 "category": "safety | reasoning | retrieval"
}
```

Minimum:

- 15–20 test cases across:

Category	Examples
Normal	Basic queries
Edge cases	Ambiguous inputs
Adversarial	Prompt injection
Safety	Harmful requests

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3. Evaluation Engine

Must include:

A. LLM-as-a-Judge

- Evaluate:
    - Correctness
    - Relevance
    - Safety

B. Rule-based checks

- Example:
    - Refusal detection
    - Keyword filters

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4. Adversarial Testing (CRITICAL)

Your system must:

- Generate OR include adversarial inputs:
    - Prompt injection
    - Jailbreak attempts

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5. Metrics & Scoring

Define:

- Pass/fail criteria
- Aggregate score

Example:

- Safety score
- Accuracy score
- Robustness score

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6. Reporting Layer

Generate:

- Summary report:
    - Scores
    - Failures
    - Insights
- Timings
    - Median
    - Mean
    - Highest
    - Lowest

Optional:

- Dashboard

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7. Observability (Bonus)

- Log:
  - Inputs
  - Outputs
  - Evaluation scores

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8. Tech Stack (Flexible)

- Python / JS
- Any framework:
  - LangChain / LlamaIndex / Semantic Kernel / OpenAI SDK

Deliverables (Put everything in a GitHub repo)

Submit the form on the link: https://forms.gle/9iPeUBHKcdHhuSq67

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1. GitHub Repo

- Framework code
- Test suite
- Evaluation pipeline

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2. Sample Agent Integration

- Plug in:
  - Simple chatbot
  - Any other agent

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3. Report

Must include:

- Metrics definition
- Results
- Failure analysis

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4. Demo **(Video or Loom)**

Show:

- Running test suite
- Adversarial failures
- Score output

Submission Instructions

Please submit your assignment using **one of the following options**:

Option 1: Public Repository (Preferred)

- Create a **public GitHub repository**
- Ensure it includes:
  - Complete source code
  - README with setup instructions
  - Any required configs or sample data
- Share the repository link in your submission

Option 2: Private Repository

If you prefer to keep your repository private:

- Add the following GitHub user as a collaborator:
  - **uptiq-chaitanya**
- Ensure:
  - The collaborator has **read access**
  - The repository is accessible before submission

Important Notes

- Incomplete or inaccessible repositories may lead to disqualification
- Ensure your code runs with the instructions provided in the README
- Include any required environment setup steps

Recommended Structure

```None
repo/
├── src/
├── data/
├── evaluation/
├── README.md
├── ARCHITECTURE.md
├── ANY_OTHER_DIAGRAMS.md
├── DEMO_VIDEO
├── requirements.txt / package.json
```